

Year 7 · Chapter 3 · Whole numbers: answers (teacher copy)

3.01 Rounding and estimation (booklet p. 3)

WE DO examples (teacher)

- Column 1, E1: 7 319 to the nearest 100 → 7 300
- Column 1, E2: 89 310 to the nearest 1 000 → 89 000
- Column 2, E1: $450 \times 12 \rightarrow \approx 500 \times 10; \approx 5\,000$
- Column 2, E2: $86 + 27 \rightarrow \approx 90 + 30; \approx 120$
- Column 3, E1: 29 students each pay \$16 for an excursion. Estimate the total. → $\approx 30 \times 20; \approx \600
- Column 3, E2: 495 people share 36 buses. About how many are on each bus? → $\approx 500 \div 40; \approx 13$

Questions (in order: Easy, Medium, Challenging)

- | | | |
|-----------|-----------------------|------------------------|
| 1. 4 200 | 18. 89 000 | 35. $\approx 20\,000$ |
| 2. 310 | 19. 180 | 36. ≈ 980 |
| 3. 7 000 | 20. $\approx 32\,000$ | 37. $\approx 15\,000$ |
| 4. 4 500 | 21. ≈ 80 | 38. $\approx 20\,000$ |
| 5. 280 | 22. $\approx 15\,000$ | 39. ≈ 110 |
| 6. 2 400 | 23. $\approx 1\,100$ | 40. $\approx 80\,000$ |
| 7. 30 000 | 24. $\approx 5\,600$ | 41. ≈ 670 |
| 8. 640 | 25. ≈ 730 | 42. $\approx \$800$ |
| 9. 4 500 | 26. ≈ 110 | 43. ≈ 23 |
| 10. 920 | 27. $\approx 1\,200$ | 44. $\approx \$1\,200$ |
| 11. 4 400 | 28. ≈ 620 | 45. ≈ 27 |
| 12. 4 900 | 29. ≈ 220 | 46. $\approx \$400$ |
| 13. 690 | 30. $\approx 63\,000$ | 47. ≈ 20 |
| 14. 6 700 | 31. $\approx 24\,000$ | 48. $\approx \$800$ |
| 15. 950 | 32. ≈ 160 | 49. ≈ 30 |
| 16. 4 500 | 33. ≈ 220 | |
| 17. 10 | 34. $\approx 6\,000$ | |

3.02 Multiplication (booklet p. 4)

WE DO examples (teacher)

- Column 1, E1: $10 \times 10 \rightarrow 100$
- Column 1, E2: $12 \times 9 \rightarrow 108$
- Column 1, E3: $67 \times 100 \rightarrow 6\,700$
- Column 2, E1: $54 \times 4 \rightarrow = 50 \times 4 + 4 \times 4; = 200 + 16; = 216$
- Column 2, E2: $33 \times 5 \rightarrow = 30 \times 5 + 3 \times 5; = 150 + 15; = 165$
- Column 3, E1: $300 \times 15 \rightarrow 300 \times 10 = 3\,000; 300 \times 5 = 1\,500; 3\,000 + 1\,500 = 4\,500$
- Column 3, E2: $326 \times 23 \rightarrow 326 \times 20 = 6\,520; 326 \times 3 = 978; 6\,520 + 978 = 7\,498$

Questions (in order: Easy, Medium, Challenging)

- | | | |
|------------|------------|------------|
| 1. 50 | 17. 3 100 | 33. 184 |
| 2. 72 | 18. 75 000 | 34. 632 |
| 3. 55 | 19. 71 000 | 35. 296 |
| 4. 18 | 20. 880 | 36. 51 |
| 5. 54 | 21. 560 | 37. 78 |
| 6. 30 | 22. 310 | 38. 144 |
| 7. 45 | 23. 9 500 | 39. 380 |
| 8. 12 | 24. 32 000 | 40. 192 |
| 9. 54 | 25. 1 200 | 41. 208 |
| 10. 84 | 26. 132 | 42. 7 392 |
| 11. 44 | 27. 704 | 43. 15 844 |
| 12. 80 | 28. 351 | 44. 6 225 |
| 13. 55 000 | 29. 450 | 45. 9 438 |
| 14. 520 | 30. 222 | 46. 5 338 |
| 15. 550 | 31. 147 | 47. 3 302 |
| 16. 240 | 32. 228 | 48. 2 074 |

3.03 Division (booklet p. 5)

WE DO examples (teacher)

- Column 1, E1: $77 \div 7 \rightarrow 11$
- Column 1, E2: $36 \div 4 \rightarrow 9$
- Column 1, E3: $120 \div 10 \rightarrow 12$
- Column 2, E1: $108 \div 6 \rightarrow = 60 \div 6 + 48 \div 6; = 10 + 8; = 18$
- Column 2, E2: $342 \div 6 \rightarrow = 300 \div 6 + 42 \div 6; = 50 + 7; = 57$
- Column 3, E1: 276 lollies are shared by 5 people. How many each, and how many are left over? → $276 \div 5 = 55 \text{ r } 1; 55 \text{ each, } 1 \text{ left over}$
- Column 3, E2: 136 lollies are shared by 9 people. How many each, and how many are left over? → $136 \div 9 = 15 \text{ r } 1; 15 \text{ each, } 1 \text{ left over}$

Questions (in order: Easy, Medium, Challenging)

- | | | |
|--------|--------|---------------------|
| 1. 5 | 17. 78 | 33. 83 |
| 2. 9 | 18. 10 | 34. 37 |
| 3. 9 | 19. 49 | 35. 91 |
| 4. 7 | 20. 71 | 36. 21 |
| 5. 10 | 21. 83 | 37. 13 |
| 6. 11 | 22. 93 | 38. 62 |
| 7. 3 | 23. 17 | 39. 66 |
| 8. 11 | 24. 95 | 40. 87 |
| 9. 6 | 25. 22 | 41. 23 |
| 10. 9 | 26. 9 | 42. 51 |
| 11. 12 | 27. 86 | 43. 17 each, 4 left |
| 12. 8 | 28. 66 | 44. 82 each, 3 left |
| 13. 6 | 29. 43 | 45. 46 each, 1 left |
| 14. 23 | 30. 66 | 46. 35 each, 1 left |
| 15. 88 | 31. 99 | 47. 66 each, 1 left |
| 16. 50 | 32. 82 | 48. 59 each, 5 left |

3.04 Divisibility tests (booklet p. 6)

WE DO examples (teacher)

- Column 1, E1: 2 298 (10) → no
- Column 1, E2: 6 988 (2) → yes
- Column 1, E3: 7 427 → 20, no
- Column 2, E1: Is 5 211 divisible by 9? → digit sum: $5 + 2 + 1 + 1 = 9$; 9 is divisible by 9, so yes
- Column 2, E2: Is 7 260 divisible by 4? → last two digits: 60; 60 is divisible by 4, so yes
- Column 2, E3: Is 5 978 divisible by 6? → even? yes; digit sum 29; no: not divisible by both 2 and 3
- Column 3, E1: $127\square$ is divisible by 3. Find every digit that fits the box. → $1 + 2 + 7 = 10$; $10 + \square$ must be a multiple of 3; $\square = 2, 5, 8$
- Column 3, E2: $651\square$ is divisible by 9. Find every digit that fits the box. → $6 + 5 + 1 = 12$; $12 + \square$ must be a multiple of 9; $\square = 6$

Questions (in order: Easy, Medium, Challenging)

- | | | |
|-------------|-------------|-------------|
| 1. no | 14. 5, no | 27. no |
| 2. no | 15. 10, no | 28. no |
| 3. no | 16. 29, no | 29. no |
| 4. no | 17. 17, no | 30. no |
| 5. yes | 18. 15, yes | 31. yes |
| 6. no | 19. 13, no | 32. no |
| 7. no | 20. 22, no | 33. no |
| 8. no | 21. 14, no | 34. yes |
| 9. no | 22. 15, yes | 35. 1, 4, 7 |
| 10. no | 23. 17, no | 36. 2, 5, 8 |
| 11. yes | 24. 22, no | 37. 0, 9 |
| 12. no | 25. 19, no | 38. 2 |
| 13. 15, yes | 26. no | 39. 6 |

3.05 Long division (booklet p. 7)

WE DO examples (teacher)

- Column 1, E1: $22 \times 7 \rightarrow 154$
- Column 1, E2: $20 \times 9 \rightarrow 180$
- Column 1, E3: 15 into 35 $\rightarrow 2 \text{ r } 5$
- Column 2, E1: $504 \div 21 \rightarrow 50 \div 21 = 2 \text{ r } 8$; bring down: $84 \div 21 = 4$; answer: 24
- Column 2, E2: $1\ 050 \div 25 \rightarrow 105 \div 25 = 4 \text{ r } 5$; bring down: $50 \div 25 = 2$; answer: 42
- Column 3, E1: 1211 students are put into teams of 33. How many full teams, and how many students are left over? $\rightarrow 1211 \div 33 = 36 \text{ r } 23$; 36 full teams, 23 left over
- Column 3, E2: 1015 students are put into teams of 36. How many full teams, and how many students are left over? $\rightarrow 1015 \div 36 = 28 \text{ r } 7$; 28 full teams, 7 left over

Questions (in order: Easy, Medium, Challenging)

- | | | |
|-----------|------------|----------------------------|
| 1. 96 | 14. 9 r 9 | 27. 24 |
| 2. 88 | 15. 4 r 3 | 28. 57 |
| 3. 66 | 16. 2 r 6 | 29. 37 |
| 4. 115 | 17. 8 r 19 | 30. 26 |
| 5. 200 | 18. 9 r 7 | 31. 33 |
| 6. 38 | 19. 7 r 4 | 32. 60 |
| 7. 52 | 20. 9 r 0 | 33. 14 teams, 35 left over |
| 8. 72 | 21. 3 r 14 | 34. 27 teams, 1 left over |
| 9. 136 | 22. 7 r 15 | 35. 20 teams, 2 left over |
| 10. 48 | 23. 4 r 7 | 36. 33 teams, 40 left over |
| 11. 84 | 24. 4 r 3 | |
| 12. 126 | 25. 4 r 12 | |
| 13. 6 r 2 | 26. 67 | |

3.06 Roman numerals (booklet p. 8)

WE DO examples (teacher)

- Column 1, E1: XXIII $\rightarrow 23$
- Column 1, E2: XL $\rightarrow 40$
- Column 1, E3: 27 $\rightarrow$ XXVII
- Column 2, E1: $490 \rightarrow 400 + 90$; CD + XC = CDXC
- Column 2, E2: $652 \rightarrow 600 + 50 + 2$; DC + L + II = DCLII
- Column 3, E1: the year 1849 $\rightarrow 1000 + 800 + 40 + 9$; M + DCCC + XL + IX; = MDCCCXLIX
- Column 3, E2: the year 2003 $\rightarrow 2000 + 3$; MM + III; = MMIII

Questions (in order: Easy, Medium, Challenging)

- | | | |
|----------|---------------|---------------|
| 1. 35 | 15. LXVII | 29. DCCCXXI |
| 2. 30 | 16. II | 30. CCCXIX |
| 3. 62 | 17. XXXV | 31. CCXCVI |
| 4. 9 | 18. XXXII | 32. CCLVIII |
| 5. 54 | 19. LXXXI | 33. CDLV |
| 6. 87 | 20. XLI | 34. CDIX |
| 7. 56 | 21. LIII | 35. CDLXXVIII |
| 8. 70 | 22. XIV | 36. MCMVI |
| 9. 98 | 23. LXXVII | 37. MCMXCIII |
| 10. 64 | 24. L | 38. MDCCCLXI |
| 11. 75 | 25. XX | 39. MCMXLIX |
| 12. 36 | 26. DCCCXLIII | 40. MCMLVI |
| 13. XLII | 27. DCXCIV | 41. MCMXV |
| 14. XXIV | 28. DCCLXXX | 42. MCMXXXVI |

3.07 Index notation and powers (booklet p. 9)

WE DO examples (teacher)

- Column 1, E1: $5 \times 5 \rightarrow 5^2$
- Column 1, E2: $9 \times 9 \rightarrow 9^2$
- Column 1, E3: $8^2 \rightarrow 64$
- Column 2, E1: $5^3 \rightarrow 5 \times 5 \times 5 = 125$

5. Column 2, E2: $4^3 \rightarrow 4 \times 4 \times 4 = 64$

6. Column 3, E1: $5^2 \times 5^2 \rightarrow 25 \times 25 = 625$

7. Column 3, E2: $4^2 \times 2^2 \rightarrow 16 \times 4 = 64$

Questions (in order: Easy, Medium, Challenging)

- | | | |
|-----------|---------|-----------|
| 1. 4^2 | 13. 49 | 28. 81 |
| 2. 2^5 | 14. 32 | 29. 16 |
| 3. 4^3 | 15. 8 | 30. 27 |
| 4. 7^2 | 16. 25 | 31. 243 |
| 5. 4^6 | 17. 4 | 32. 32 |
| 6. 7^3 | 18. 81 | 33. 34 |
| 7. 2^4 | 19. 64 | 34. 20 |
| 8. 7^6 | 20. 100 | 35. 2 000 |
| 9. 3^3 | 21. 125 | 36. 13 |
| 10. 3^4 | 22. 81 | 37. 89 |
| 11. 6^3 | 23. 16 | 38. 43 |
| 12. 2^3 | 24. 243 | 39. 100 |
| | 25. 36 | 40. 225 |
| | 26. 8 | 41. 32 |
| | 27. 216 | 42. 36 |

3.08 Square roots and cube roots (booklet p. 10)

10)

WE DO examples (teacher)

- Column 1, E1: $\sqrt{49} \rightarrow 7$
- Column 1, E2: $\sqrt{100} \rightarrow 10$
- Column 1, E3: $\sqrt[3]{8} \rightarrow 2$
- Column 2, E1: $\sqrt{47} \rightarrow 6^2 = 36, 7^2 = 49$; between 6 and 7
- Column 2, E2: $\sqrt{89} \rightarrow 9^2 = 81, 10^2 = 100$; between 9 and 10
- Column 3, E1: $\sqrt{121} + \sqrt[3]{125} \times 4 \rightarrow 11 + 5 \times 4 = 11 + 20 = 31$
- Column 3, E2: $\sqrt{100} + \sqrt[3]{64} \times 2 \rightarrow 10 + 4 \times 2 = 10 + 8 = 18$

Questions (in order: Easy, Medium, Challenging)

- | | | |
|--------|---------------|---------------|
| 1. 13 | 14. 4 | 27. 7 and 8 |
| 2. 2 | 15. 3 | 28. 5 and 6 |
| 3. 14 | 16. 10 | 29. 9 and 10 |
| 4. 15 | 17. 5 | 30. 3 and 4 |
| 5. 5 | 18. 6 | 31. 11 and 12 |
| 6. 18 | 19. 9 | 32. 11 and 12 |
| 7. 19 | 20. 7 | 33. 32 |
| 8. 20 | 21. 8 | 34. 11 |
| 9. 17 | 22. 1 | 35. 22 |
| 10. 12 | 23. 11 and 12 | 36. 6 |
| 11. 9 | 24. 6 and 7 | 37. 36 |
| 12. 3 | 25. 11 and 12 | 38. 24 |
| 13. 6 | 26. 5 and 6 | 39. 25 |

3.09 Prime and composite numbers (booklet p. 11)

p. 11)

WE DO examples (teacher)

- Column 1, E1: 80 $\rightarrow$ C
- Column 1, E2: 19 $\rightarrow$ P
- Column 2, E1: $52 \rightarrow 1 \times 52, 2 \times 26, 4 \times 13$; factors: 1, 2, 4, 13, 26, 52
- Column 2, E2: $38 \rightarrow 1 \times 38, 2 \times 19$; factors: 1, 2, 19, 38
- Column 3, E1: Is 63 prime? $\rightarrow 63 \div 3 = 21$; composite: $63 = 3 \times 21$
- Column 3, E2: Is 55 prime? $\rightarrow 55 \div 5 = 11$; composite: $55 = 5 \times 11$

Questions (in order: Easy, Medium, Challenging)

- | | | |
|------|------|-------|
| 1. P | 5. C | 9. C |
| 2. C | 6. C | 10. C |
| 3. C | 7. C | 11. C |
| 4. C | 8. C | 12. C |

- | | | |
|-------|-------------------------|-----------------------------|
| 13. C | 26. C | 38. 1, 2, 3, 4, 6, 8, |
| 14. C | 27. C | 12, 24 |
| 15. C | 28. C | 39. 1, 2, 4, 8, 16, 32 |
| 16. P | 29. C | 40. no, $77 = 7 \times 11$ |
| 17. C | 30. 1, 2, 3, 5, 6, 10, | 41. no, $75 = 3 \times 25$ |
| 18. C | 15, 30 | 42. yes |
| 19. P | 31. 1, 2, 23, 46 | 43. no, $93 = 3 \times 31$ |
| 20. C | 32. 1, 3, 7, 21 | 44. yes |
| 21. C | 33. 1, 2, 5, 10, 25, 50 | 45. yes |
| 22. C | 34. 1, 2, 17, 34 | 46. yes |
| 23. C | 35. 1, 2, 3, 4, 6, 12 | 47. yes |
| 24. C | 36. 1, 2, 4, 11, 22, | 48. no, $105 = 3 \times 35$ |
| 25. P | 44 | 49. no, $135 = 3 \times 45$ |
| | 37. 1, 2, 4, 5, 10, 20 | |

3.10 Prime factors (booklet p. 12)

WE DO examples (teacher)

- Column 1, E1: $49 \rightarrow 7 \times 7$
- Column 1, E2: $65 \rightarrow 5 \times 13$
- Column 2, E1: $40 \rightarrow 40 \div 2 = 20, 20 \div 2 = 10; 10 \div 2 = 5, \dots; 40 = 2^3 \times 5$
- Column 2, E2: $36 \rightarrow 36 \div 2 = 18, 18 \div 2 = 9; 9 \div 3 = 3, \dots; 36 = 2^2 \times 3^2$
- Column 3, E1: $\sqrt{36} \rightarrow 36 = 2^2 \times 3^2; = (2 \times 3)^2; \sqrt{36} = 6$
- Column 3, E2: $\sqrt[3]{216} \rightarrow 216 = 2^3 \times 3^3; = (2 \times 3)^3; \sqrt[3]{216} = 6$

Questions (in order: Easy, Medium, Challenging)

- | | | |
|-------------------|---------------------------|-----------------------------|
| 1. 11×13 | 12. 7×13 | 23. $2 \times 3 \times 5$ |
| 2. 3×13 | 13. 5×5 | 24. $2^2 \times 5$ |
| 3. 7×11 | 14. 2×11 | 25. $2^2 \times 3 \times 5$ |
| 4. 2×7 | 15. 2×2 | 26. 2×3^2 |
| 5. 13×13 | 16. 2×3 | 27. 18 |
| 6. 11×11 | 17. 2×13 | 28. 10 |
| 7. 5×11 | 18. 3×3 | 29. 30 |
| 8. 5×7 | 19. 3×11 | 30. 12 |
| 9. 2×5 | 20. $2^3 \times 3$ | 31. 22 |
| 10. 3×5 | 21. $2 \times 3 \times 7$ | 32. 15 |
| 11. 3×7 | 22. $2 \times 5 \times 7$ | 33. 15 |

3.11 Highest common factor (booklet p. 13)

WE DO examples (teacher)

- Column 1, E1: 8 and 10 $\rightarrow 2$
- Column 1, E2: 24 and 20 $\rightarrow 4$
- Column 2, E1: 16 and 32 $\rightarrow 16$: 1, 2, 4, 8, 16; 32: 1, 2, 4, 8, 16, 32; HCF = 16
- Column 2, E2: 16 and 24 $\rightarrow 16$: 1, 2, 4, 8, 16; 24: 1, 2, 3, 4, 6, 8, 12, 24; HCF = 8
- Column 3, E1: 30 and 75 $\rightarrow 30 = 2 \times 3 \times 5; 75 = 3 \times 5 \times 5; \text{HCF} = 3 \times 5 = 15$

6. Column 3, E2: 40 and 50 $\rightarrow 40 = 2 \times 2 \times 2 \times 5; 50 = 2 \times 5 \times 5; \text{HCF} = 2 \times 5 = 10$

Questions (in order: Easy, Medium, Challenging)

- | | | |
|-------|-------|--------|
| 1. 9 | 16. 2 | 31. 5 |
| 2. 12 | 17. 6 | 32. 4 |
| 3. 3 | 18. 4 | 33. 2 |
| 4. 2 | 19. 6 | 34. 2 |
| 5. 9 | 20. 2 | 35. 6 |
| 6. 5 | 21. 2 | 36. 3 |
| 7. 2 | 22. 2 | 37. 12 |
| 8. 9 | 23. 3 | 38. 10 |
| 9. 7 | 24. 3 | 39. 18 |
| 10. 4 | 25. 8 | 40. 10 |
| 11. 6 | 26. 2 | 41. 15 |
| 12. 8 | 27. 9 | 42. 15 |
| 13. 6 | 28. 8 | 43. 15 |
| 14. 4 | 29. 8 | |
| 15. 4 | 30. 2 | |

3.12 Lowest common multiple (booklet p. 14)

WE DO examples (teacher)

- Column 1, E1: 10 and 5 $\rightarrow 10$
- Column 1, E2: 11 and 8 $\rightarrow 88$
- Column 2, E1: 3 and 12 $\rightarrow 3$: 3, 6, 9, 12; 12: 12; LCM = 12
- Column 2, E2: 6 and 5 $\rightarrow 6$: 6, 12, 18, 24, 30; 5: 5, 10, 15, 20, 25, 30; LCM = 30
- Column 3, E1: One bus leaves every 12 minutes, another every 7 minutes. Both leave at 8:00. After how many minutes do they next leave together? $\rightarrow \text{LCM of 12 and 7} = 84$; after 84 minutes
- Column 3, E2: One bus leaves every 7 minutes, another every 11 minutes. Both leave at 8:00. After how many minutes do they next leave together? $\rightarrow \text{LCM of 7 and 11} = 77$; after 77 minutes

Questions (in order: Easy, Medium, Challenging)

- | | | |
|--------|--------|----------------|
| 1. 63 | 14. 20 | 27. 88 |
| 2. 9 | 15. 66 | 28. 12 |
| 3. 132 | 16. 72 | 29. 24 |
| 4. 55 | 17. 15 | 30. 6 |
| 5. 24 | 18. 8 | 31. 12 |
| 6. 8 | 19. 18 | 32. 12 |
| 7. 66 | 20. 14 | 33. 12 |
| 8. 42 | 21. 33 | 34. 12 |
| 9. 14 | 22. 12 | 35. 40 |
| 10. 30 | 23. 8 | 36. 24 |
| 11. 22 | 24. 56 | 37. 70 minutes |
| 12. 20 | 25. 28 | 38. 42 minutes |
| 13. 24 | 26. 6 | 39. 10 minutes |